

COMPARISON OF FIXED POINT THEOREMS

THEOREM	SPACE TYPE	CONDITION	CONSTRUCTIVE	RATE	UNIQUENESS
Banach Contraction	Complete Metric	Contraction: $d(Tx,Ty) \leq \alpha d(x,y)$ $\alpha < 1$	YES (Picard) Algorithm available	Linear (α^n)	YES
Brouwer's FPT	Compact Convex Set in $\mathbb{R}^n$	Continuous mapping of X into itself	NO (exists but no algorithm)	No info	Not always
Schauder's FPT	Banach Space	Continuous + Compact image	NO construction	No info	Not always
Kakutani's FPT (Set-valued)	Compact Convex Set	Upper semi- continuous multifunctions	NO (for multi- functions)	No info	Not always

KEY DIFFERENCES:

- BANACH: Requires CONTRACTION property → guarantees UNIQUE fixed point
Provides EXPLICIT ALGORITHM with ERROR BOUNDS
Works in any COMPLETE METRIC SPACE (including infinite dimensions)
- BROUWER: Only requires CONTINUITY → may have multiple fixed points
Existence only, NO ALGORITHM
Limited to COMPACT CONVEX subsets of $\mathbb{R}^n$
- SCHAUDER: Generalizes Brouwer to infinite dimensions
Existence only, NO ALGORITHM
Requires COMPACTNESS condition on the mapping

CHOICE OF THEOREM:

- ✓ Use BANACH when: Contraction property is satisfied, uniqueness matters, efficient algorithm needed
- ✓ Use BROUWER when: Finite dimensions, continuity sufficient, domain is compact convex subset
- ✓ Use SCHAUDER when: Infinite dimensions, compactness is available, existence is sufficient